

FinSage-7B — Benchmark Report

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> ■■ ****Sample/mock report for pipeline validation only. Not real benchmark results.****

>

> This report was generated because the underlying evaluation artifacts were produced by the ****mock**** generator (no real fine-tuned weights). The numbers below validate the reporting pipeline only and ****must not be published as real results****. Re-run after a real fine-tune + evaluation to produce a publishable report.

Executive Summary

Across ****8**** overall metrics, FinSage-7B improved on ****5**** and regressed on ****2**** versus the base Mistral-7B-Instruct model, with a mean absolute delta of ****0.176****.

FinSage-7B is a QLoRA fine-tune of Mistral-7B-Instruct specialised for analysing U.S. SEC filings (10-K / 10-Q / 8-K). It is designed to stay grounded in the supplied filing text, extract concrete metrics, and avoid investment advice. This report documents the data, training, evaluation methodology, and the measured base-vs-fine-tuned comparison.

Project Overview

FinSage-7B is an end-to-end, reproducible pipeline: SEC EDGAR ingestion → section extraction → instruction-dataset construction → baseline evaluation → QLoRA fine-tuning → fine-tuned evaluation → vLLM serving → FastAPI wrapper → web demo → Docker deployment. Every stage is covered by tests and runs on commodity hardware (training on a single rented GPU; everything else CPU-only).

Why Financial Filings

SEC filings are long, dense, and high-stakes: analysts spend hours locating risk factors, MD&A drivers, and reported metrics. They are also public and well-structured, which makes them an ideal domain for demonstrating that a small, cheaply fine-tuned model can be made ****more grounded and specific**** than its base model on a real professional task — without hallucinating numbers or drifting into advice.

Dataset Summary

Dataset statistics not available.

Training Setup

Training summary not available (model not yet fine-tuned).

Evaluation Methodology

Both models are evaluated on the same held-out instruction set with identical prompts. Metrics include exact match, token-level F1, ROUGE-L, numeric precision/recall/exact-match (for extraction tasks), classification accuracy (for outlook/hallucination tasks), and a faithfulness score (lexical overlap by default; optional NLI entailment). The base model uses the same generation settings as the fine-tuned model so the comparison is apples-to-apples.

Overall Results

Metric	Base	Fine-tuned	Δ abs	Δ %	Result
classification accuracy	0.500	0.500	+0.000	+0.0	— no change
exact match	0.100	0.000	-0.100	-100.0	▼ regressed
lexical faithfulness	0.700	0.896	+0.196	+28.0	▲ improved
numeric exact match	0.250	0.000	-0.250	-100.0	▼ regressed
numeric precision	0.400	1.000	+0.600	+150.0	▲ improved
numeric recall	0.350	0.750	+0.400	+114.3	▲ improved
rouge l	0.300	0.645	+0.345	+115.0	▲ improved
token f1	0.420	0.640	+0.220	+52.3	▲ improved

![Overall metrics: base vs fine-tuned](figures/report_overall_metrics.png)

Task-wise Results

Task	Metric	Base	Fine-tuned	Δ abs	Result
analyst summary	lexical faithfulness	0.000	1.000	+1.000	▲ improved
analyst summary	rouge l	0.000	0.400	+0.400	▲ improved
analyst summary	token f1	0.000	0.533	+0.533	▲ improved
business risk identification	lexical faithfulness	0.000	1.000	+1.000	▲ improved
business risk identification	rouge l	0.000	0.846	+0.846	▲ improved
business risk identification	token f1	0.000	0.846	+0.846	▲ improved
filing qa	exact match	0.000	0.000	+0.000	— no change
filing qa	lexical faithfulness	0.720	1.000	+0.280	▲ improved
filing qa	token f1	0.400	0.800	+0.400	▲ improved

hallucination detection	classification accuracy	0.500	1.000	+0.500	▲ improved
hallucination detection	lexical faithfulness	0.600	0.400	-0.200	▼ regressed
mda explanation	lexical faithfulness	0.000	1.000	+1.000	▲ improved
mda explanation	rouge l	0.000	0.171	+0.171	▲ improved
mda explanation	token f1	0.000	0.343	+0.343	▲ improved
metric extraction	lexical faithfulness	0.660	0.667	+0.007	▲ improved
metric extraction	numeric exact match	0.250	0.000	-0.250	▼ regressed
metric extraction	numeric precision	0.000	1.000	+1.000	▲ improved
metric extraction	numeric recall	0.000	0.750	+0.750	▲ improved
metric extraction	token f1	0.500	0.667	+0.167	▲ improved
outlook classification	classification accuracy	0.500	0.000	-0.500	▼ regressed
outlook classification	token f1	0.300	0.118	-0.182	▼ regressed
revenue driver explanation	lexical faithfulness	0.000	1.000	+1.000	▲ improved
revenue driver explanation	rouge l	0.000	1.000	+1.000	▲ improved
revenue driver explanation	token f1	0.000	1.000	+1.000	▲ improved
risk summary	lexical faithfulness	0.740	1.000	+0.260	▲ improved
risk summary	rouge l	0.310	0.452	+0.142	▲ improved
risk summary	token f1	0.450	0.452	+0.002	▲ improved
yoy comparison	lexical faithfulness	0.000	1.000	+1.000	▲ improved
yoy comparison	rouge l	0.000	1.000	+1.000	▲ improved
yoy comparison	token f1	0.000	1.000	+1.000	▲ improved

![[Mean absolute delta by task](figures/report_task_delta.png)]

Hallucination and Faithfulness Analysis

Faithfulness measures whether the answer stays grounded in the provided excerpt. The chart below contrasts faithfulness-related metrics for the base and fine-tuned models; higher is better. Note the default faithfulness metric is a lexical proxy, not a full entailment audit (see Limitations).

![[Faithfulness metrics](figures/report_hallucination.png)]

Latency and Deployment Summary

Latency benchmark not available.

The serving stack is a public CPU-only FastAPI wrapper (auth, rate limiting, structured logging, financial-disclaimer injection) in front of an internal, GPU-bound vLLM OpenAI-compatible server. The

full stack is packaged with Docker Compose (production, demo, and GPU overlays).

Qualitative Examples

Example 1 — Improvement (revenue driver explanation)

Instruction: Identify revenue drivers.

Filing excerpt: Recurring subscription revenue and cloud growth are the primary drivers.

Reference answer: - Recurring subscription revenue and cloud growth are the primary drivers.

Base model: Subscription revenue is a driver.

Fine-tuned (FinSage-7B): Recurring subscription revenue and growth in the services segment are the primary revenue drivers, supported by pricing and customer demand.

Metric change: lexical faithfulness: +0.220, rouge l: +0.620, token f1: +0.580

Takeaway: Fine-tuned answer is more specific and better grounded in the excerpt.

Example 2 — Improvement (yoy comparison)

Instruction: Identify YoY comparison.

Filing excerpt: Services revenue increased 14% year over year.

Reference answer: - Services revenue increased 14% year over year.

Base model: Services revenue increased.

Fine-tuned (FinSage-7B): Services revenue increased 14% year over year while product revenue grew 6% compared with the prior year.

Metric change: lexical faithfulness: +0.250, rouge l: +0.600, token f1: +0.500

Takeaway: Fine-tuned answer is more specific and better grounded in the excerpt.

Example 3 — Regression / failure case (outlook classification)

Instruction: Classify the outlook.

Filing excerpt: Management expects continued growth and strong demand.

Reference answer: {"label": "positive"}

Base model: The outlook is neutral.

Fine-tuned (FinSage-7B): The outlook is neutral based on the excerpt.

Metric change: classification accuracy: +0.000, token f1: -0.232

****Takeaway:**** Fine-tuning regressed on lexical-overlap metrics here, often because the fine-tuned answer adds correct detail not present in the short reference.

Example 4 — Average case (analyst summary)

****Instruction:**** Write an analyst summary.

****Filing excerpt:**** Strong demand and expanding margins offset supply chain risks.

****Reference answer:**** Analyst Summary: Strong demand and margins offset by risks.

****Base model:**** Strong demand and margins.

****Fine-tuned (FinSage-7B):**** The company reported strong demand and expanding margins but flagged supply chain and competition risks for the coming year.

****Metric change:**** lexical faithfulness: +0.280, rouge l: +0.100, token f1: +0.133

****Takeaway:**** Fine-tuned answer is more specific and better grounded in the excerpt.

Example 5 — Faithfulness / hallucination case (metric extraction)

****Instruction:**** Extract metrics.

****Filing excerpt:**** Net revenue was \$81,462 million, up 51%.

****Reference answer:**** - \$81,462 million

- 51%

****Base model:**** Reported metrics: \$81,462 million.

****Fine-tuned (FinSage-7B):**** Reported metrics: \$81,462, 51%, 25.6%, \$12,556.

****Metric change:**** lexical faithfulness: +0.007, numeric exact match: +0.000, numeric precision: +1.000, numeric recall: +0.750, token f1: +0.167

****Takeaway:**** Fine-tuning improved grounding/faithfulness relative to the base model.

Error Analysis and Regressions

At least one task regressed on lexical-overlap metrics. In manual review this is usually because the fine-tuned model adds correct, filing-grounded detail that the short reference answer omits — penalising overlap scores while improving usefulness. This is a known weakness of n-gram metrics on open-ended generation and motivates the faithfulness/NLI track.

Limitations

| Limitation | Detail |

| --- | --- |

| Weak-supervision targets | Phase 3 instruction targets are template/extractive (no LLM teacher), so reference answers approximate, not certify, ground truth. |

| Lexical faithfulness proxy | The default faithfulness metric is lexical-overlap based; optional NLI entailment is available but off by default. Neither is a full hallucination audit. |

| Small evaluation set | The held-out evaluation set is small; metric deltas have wide confidence intervals and should be read as directional, not definitive. |

| Single base model | Only Mistral-7B-Instruct is compared; results may not transfer to other base models or larger scales. |

| No live market data | The model reasons only over the provided filing excerpt; it has no access to prices, real-time disclosures, or post-filing events. |

| Not investment advice | Outputs are informational only and must not be used for investment decisions. |

Financial Safety Disclaimer

FinSage-7B is **not** a licensed financial advisor. Its outputs are informational summaries of the supplied text only, are **not** investment recommendations, and may be incomplete or incorrect. Always verify against the original filing and consult a qualified professional before making decisions.

Reproducibility Guide

```
```bash
```

### 1. Build the instruction dataset (no network in test mode)

```
make build-dataset && make validate-dataset
```

### 2. Baseline eval (mock backend is CPU-only; real needs a GPU)

```
make eval-baseline
```

### 3. Fine-tune (GPU) then evaluate + compare

```
make train && make eval-finetuned && make compare-models
```

### 4. Regenerate this report

```
make report && make validate-report
```

```
```
```

Appendix

See report_appendix.md for metric and task-type definitions, the dataset split strategy, evaluation caveats, the prompt format, and full reproducibility commands.

****Artifacts not available at report time:**** `api\_latency`, `dataset\_stats`, `deployment\_health`,
`training\_summary`, `validation\_report`, `vllm\_latency`.